

Technical Document #23: Retrieval Augmented Generation - Comprehensive Overview

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Retrieval-Augmented Generation (RAG) combines information retrieval with text generation. It grounds LLM responses in factual, retrieved documents.

Hybrid search combines keyword-based and semantic search. It leverages the strengths of both approaches for better retrieval quality.

Query decomposition breaks complex queries into simpler sub-queries. It improves retrieval quality for multi-faceted questions.

Evaluation metrics for RAG include faithfulness, answer relevance, and context recall. They measure the quality of the retrieval-generation pipeline.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

The rapid advancement of technology has transformed how organizations approach problem-solving and innovation. Companies must adapt to stay competitive in an ever-evolving landscape.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Key Takeaways:

- Retrieval-Augmented Generation (RAG) combines information retrieval with text generation.
- Context window optimization selects the most relevant information to fit within the LLM's context limit.
- Vector databases store dense embeddings for semantic search.